

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Electronics Devices and Circuits (New)

Semester: Spring

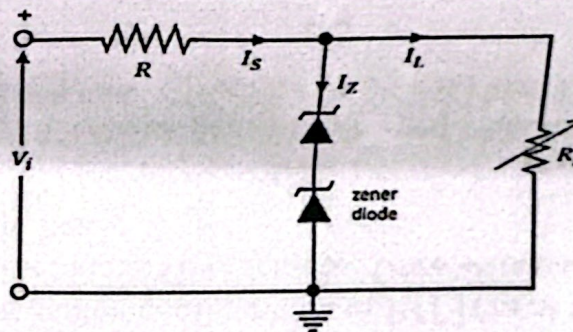
Year : 2026
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

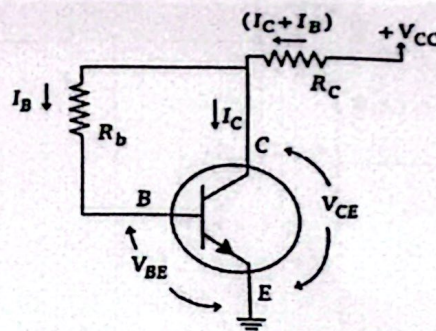
Attempt all the questions.

1. a) Explain the VI characteristics of PN Junction diode with necessary diagrams. 7
- b) A Zener regulator consist of two Zener diode with 6 V Zener voltage each, with variable load resistance from 200Ω to $1\text{ K}\Omega$. Given that below $V_{in} = 15\text{ V}$ and $R = 100\Omega$. Find V_o , I_L , I_z 8



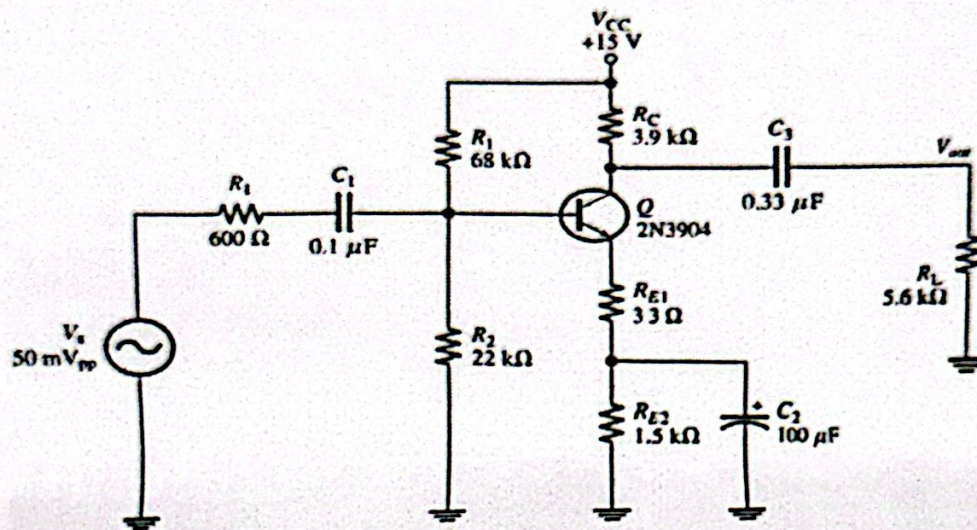
Draw the VI characteristics curve of Zener diode according to above solution.

2. a) Discuss the biasing required for various modes of operation of BJT. Find Q point or operating point (V_{CE} , I_C) for circuit given and DC load line. Here $\beta = 100$, $V_{cc} = 20\text{V}$, $R_c = 1\text{K}\Omega$, $R_b = 100\text{ K}\Omega$. 8

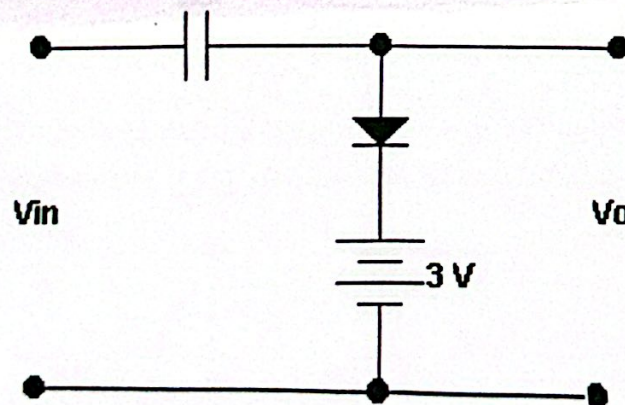


OR

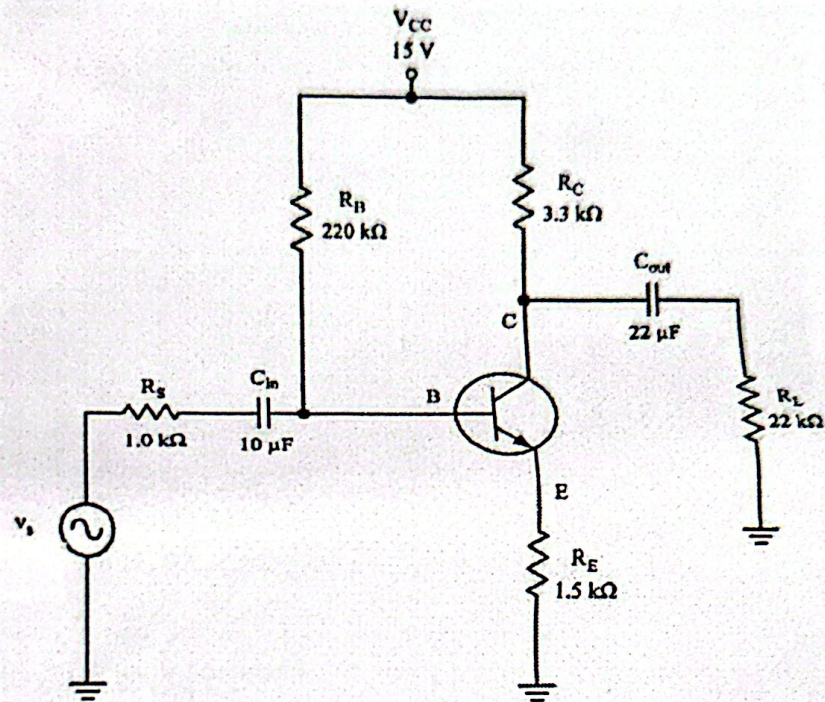
For the transistor amplifier shown, draw D.C. load line and locate operating point on it. (Assume $\beta=90$)



- b) What are the current gains in a transistor? Derive the analytical expression of collector current in a BJT. 7
3. a) Draw the output waveform for the given circuit for the input sinusoidal signal having $V_p = 10$ V. 8



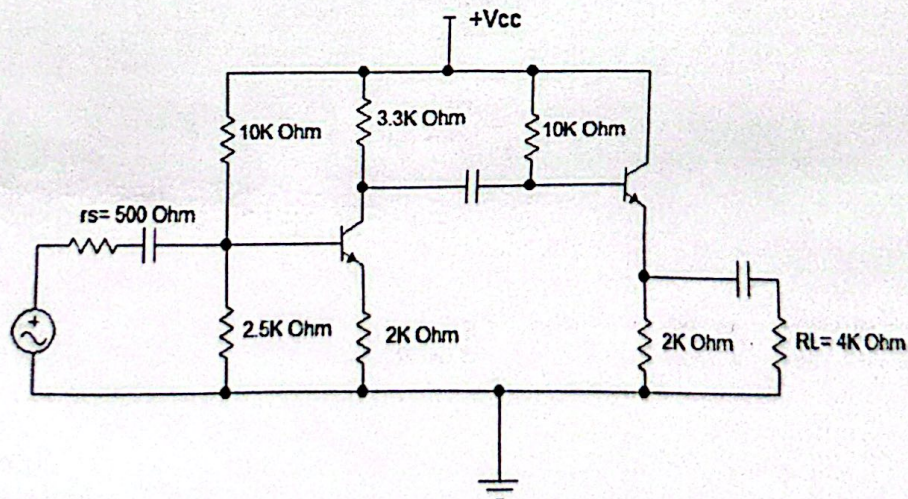
- b) Explain the working principle of full wave bridge rectifier and compare ripple factor of half wave rectifier with full wave rectifier. 7
4. a) Why JFET is called voltage-controlled device? Draw the structure and output characteristics of n-channel JFET. Indicate different region in the curve and explain its significance. 7
- b) Determine overall voltage gain using r_e model for the circuit shown below. Take $\beta_1 = 120$ and $r_e = 20$ ohm. 8



5. a) Why common emitter configuration of BJT is used in intermediate stages while cascading. Give reasons. Draw two stage RC coupled transistor amplifier. 8

OR

Find the overall voltage gain of cascaded transistor amplifier as shown in figure below. Given: $r_{e1} = r_{e2} = 25\Omega$ and $\beta_1 = \beta_2 = 110$.



- b) Explain the working of Class B amplifier with required circuit diagram and derive its efficiency. 7

6. a) What is Barkhausen criterion for Oscillator? Design a Wein-bridge oscillator to operate at a frequency of 5KHz. 8
- b) Compare the characteristics of ideal and non-ideal Op-Amp. Design an amplifier using two inverting configurations to get the output expressions as 7
- $V_o = 4V_1 - 0.5V_2 + 5V_3 - 6V_4$
- where V_1, V_2, V_3, V_4 are inputs. Assume $R_f = 52 \text{ K}\Omega$.
7. Write short notes on: (Any two) 2×5
- a) Cross over distortion and its improvement
- b) Junction capacitance
- c) Clipper and clamper circuit